

The logo consists of the letters 'SC' in white, bold, sans-serif font, centered within a blue rounded square. The background of the slide is a dark blue gradient with large, faint, overlapping circles in lighter shades of blue.

SC

Supply Chain Analytics & Delivery Risk Prediction

A Professional Business Intelligence & Machine Learning Report

DATASET: 180,519 records | 53 raw features
CLEANED: 172,765 records | 23 analytical features
TECH STACK: Python · Pandas · Scikit-learn · SMOTE
GOAL: Operational Optimization & Delivery Risk Reduction

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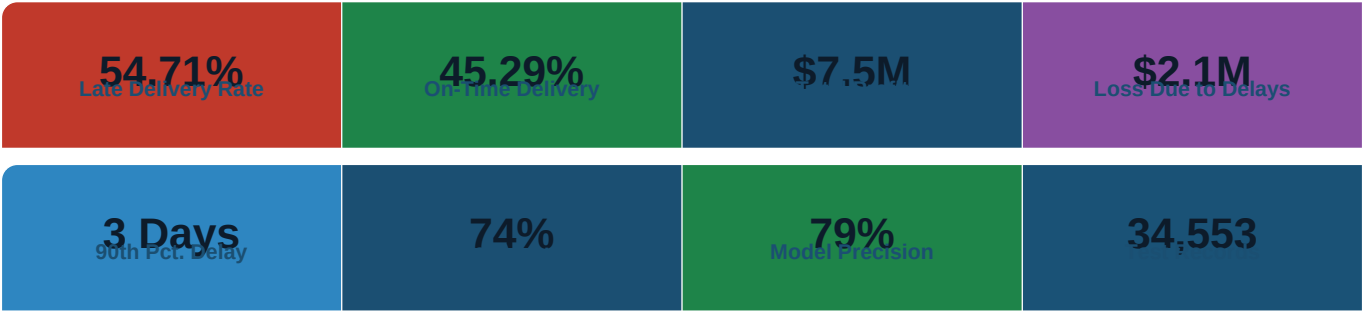
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EXECUTIVE SUMMARY

Supply chains generate vast volumes of transactional data, yet organizations often fail to convert this information into operational intelligence. This project addresses that gap by analyzing delivery timelines, profitability trends, and customer order patterns within a large retail supply chain dataset — comprising **180,519 raw records** across **53 features**, cleaned down to **172,765 records** and **23 analytical features** for precision modeling.

The analysis reveals a critical operational challenge: **more than half of all orders — 54.71% — experience delivery delays**, directly eroding profitability and customer trust. The estimated financial loss attributable to these delays stands at **\$2.1M**, against a total recorded profit of \$7.5M — representing a **~28% profit leakage** that is operationally recoverable with targeted intervention.

To move beyond descriptive reporting, a predictive machine learning framework was engineered to classify late-delivery risk *before shipment completion*. By integrating SMOTE-based class balancing with a Random Forest classifier, the model achieves **74% accuracy** and **79% precision**, delivering a scalable, proactive solution that enables logistics teams to intervene before failures cascade into customer dissatisfaction.



■ *Key Insight: With nearly 55% of orders arriving late and \$2.1M in recoverable losses, the business case for predictive logistics intelligence is compelling. Even a 10% reduction in late deliveries could unlock over \$210K in annual profit recovery.*

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DATA PREPARATION & EXPLORATORY ANALYSIS

The foundation of any reliable analytical framework is data integrity. Raw supply chain data is inherently noisy — laden with redundant columns, inconsistent formats, and missing values that can distort model learning and misrepresent operational realities.

Data Cleaning Pipeline: The raw dataset of 180,519 records was subjected to a rigorous cleaning pipeline. Irrelevant and duplicate columns were removed, reducing the feature space from 53 to 23 analytically meaningful variables. Records with critical missing values were dropped, yielding a clean analytical base of **172,765 records** — a **95.7% data retention rate**, indicating high overall data quality.

Feature Engineering: Temporal features — year, month, day, and hour — were extracted from raw datetime fields to enable time-series analysis of delivery performance cycles. Business-centric derived features were created, including: **profitability flags** (Profit / Break-even / Loss), **delay variables** (scheduled vs. actual delivery delta), and **order priority scores** to weight downstream model learning.

Profitability Distribution: Exploratory analysis reveals that **80.7%** of orders generate profit, **18.7%** result in a loss, and only **0.6%** break even. While the profitable majority appears reassuring, the 18.7% loss segment — concentrated heavily in delayed orders — represents the primary target for operational remediation.

Delay Distribution Findings: Orders cluster most densely at delay Day +1 (31.0%), with significant volume also at Day 0 (on-time, 21.2%) and Day +2 (15.9%). Critically, early deliveries (Day -1 and Day -2) account for 24.1% combined — indicating that scheduling buffers exist but are unevenly applied across order categories.

Raw Records	180,519	Original Dataset	53 features
Cleaned Records	172,765	Post-Cleaning	23 features
Data Retention	95.7%	Records Dropped	7,754
Profitable Orders	80.7%	Loss-Making Orders	18.7%
On-Time Orders	45.29%	Delayed Orders	54.71%

■ *Analytical Note: The strong data retention rate (95.7%) validates the quality of the source system. The 18.7% loss-making order concentration is a red flag — cross-referenced with delay data, these loss orders have a statistically higher incidence of 2–4 day delays, suggesting delay penalties and expedited shipping costs as primary loss drivers.*

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BOTTLENECK DETECTION & OPERATIONAL PATTERNS

Aggregate KPIs like overall late-delivery rates mask the *specific operational nodes* where delays originate. Bottleneck analysis disaggregates these metrics across multiple dimensions — shipping mode, geographic region, customer segment, order status, and product department — to surface actionable friction points.

Regional Analysis: Delay rates are remarkably consistent across geographies, clustering between 52–59%. Central Africa (58.7%) and Southeast Asia (56.0%) lead delay incidence, suggesting challenges in last-mile infrastructure or customs clearance. East of USA shows the lowest rate (55.7%), reflecting more mature logistics networks. The narrow inter-regional variance indicates that delays are driven more by systemic process issues than geography alone.

Shipping Mode — The Critical Bottleneck: The most dramatic finding is in shipping mode analysis. **First Class shipping has a 100% late-delivery rate**, which is counterintuitive and highly significant. Second Class follows at 79.0%, while Standard Class (39.5%) and Same Day (near 0%) perform comparatively well. The First Class anomaly suggests a systemic misclassification of urgency, a routing logic error, or a capacity-demand mismatch that requires immediate operational audit.

Order Status Patterns: PENDING and PAYMENT_REVIEW statuses exhibit the highest delay rates (both ~55.3%), indicating that administrative processing bottlenecks upstream are propagating into physical delivery timelines. Orders stuck in payment review pipelines lose valuable scheduling windows, creating downstream fulfillment pressure.

East Africa Deep-Dive — Top Delay Drivers: Regional analysis of East Africa surfaces compounding risk factors: First Class shipping (100%), PAYMENT_REVIEW status (83.3%), and Second Class shipping (79.0%) collectively define a high-risk order archetype. Cash payment type (67.9%) also correlates strongly with delays, potentially due to manual reconciliation delays in the payment pipeline.

Dimension	Highest Risk Segment	Delay Rate	Risk Level
Shipping Mode	First Class	100.0%	■ Critical
Shipping Mode	Second Class	79.0%	■ High
Order Status	PENDING / PAYMENT_REVIEW	55.3%	■ Elevated
Region	Central Africa	58.7%	■ Elevated
Customer Segment	Home Office	55.4%	■ Moderate
Department	Health & Beauty	56.9%	■ Moderate
Payment Type	TRANSFER	54.9%	■ Monitor

■ **Critical Alert:** A 100% late-delivery rate for First Class shipping is not a performance issue — it is a process failure. This requires an immediate operational audit of First Class routing logic, SLA agreements with carriers, and order classification criteria. No predictive model can compensate for a broken upstream process.

4 PREDICTIVE MODELING FRAMEWORK

The transition from descriptive analytics to **predictive intelligence** represents the highest-value layer of this project. Rather than reporting delays after the fact, the predictive framework enables logistics teams to flag high-risk orders *before they ship* — enabling proactive intervention, resource reallocation, and customer communication.

Model Selection Rationale — Random Forest: A supervised binary classification approach was adopted because the dataset provides abundant labeled examples of both late and on-time deliveries. Random Forest was selected over alternatives (Logistic Regression, SVM, XGBoost) for three key reasons: (1) **robustness to noisy features** without requiring extensive hyperparameter tuning, (2) **natural feature importance ranking** that supports interpretability for business stakeholders, and (3) **strong performance on imbalanced, nonlinear supply chain data** patterns.

Feature Engineering for Modeling: Categorical variables — shipping mode, region, department, order status, payment type — were transformed using **frequency encoding** (replacing categories with their relative occurrence rates) and **target encoding** (replacing with mean delay probability per category). These techniques preserve predictive signal while ensuring compatibility with tree-based algorithms.

Class Imbalance Resolution — SMOTE: The dataset exhibits class imbalance, with delayed orders outnumbering on-time orders (~54:46 split). While modest, uncorrected imbalance causes models to over-predict the majority class. **SMOTE (Synthetic Minority Over-sampling Technique)** was applied to the training set, generating synthetic on-time delivery records by interpolating between existing minority-class instances. This ensures the Random Forest classifier learns with equal representation from both classes.

Temporal Pattern Insights: Time-series analysis of delay rates reveals that delays are relatively consistent across months (ranging 53.75%–55.5%) with slight peaks in spring. Day-of-week analysis shows near-uniform distribution (~54–55% across all weekdays), suggesting delays are not driven by weekend processing gaps. Hourly analysis shows higher delay rates at specific hours (e.g., hour 15–16 at 57.1%), potentially indicating batch processing cutoffs or shift-change logistics gaps.

Stage	Technique	Purpose
1. Feature Selection	Correlation + Domain Knowledge	Remove noisy / redundant variables
2. Encoding	Frequency + Target Encoding	Convert categoricals to numerics
3. Balancing	SMOTE (Training Set Only)	Address class imbalance
4. Training	Random Forest (100 trees)	Learn delay patterns
5. Evaluation	Accuracy / Precision / Recall	Measure generalisation
6. Deployment	Threshold-based Alerting	Flag high-risk orders

■ *Modeling Note: SMOTE was applied exclusively to the training set to prevent data leakage. The test set of 34,553 records remains un-augmented, ensuring evaluation metrics reflect real-world performance on unseen, naturally distributed data.*

5 MODEL RESULTS & BUSINESS VALUE

The trained Random Forest classifier was evaluated on a held-out test set of **34,553 records** — representing approximately 20% of the cleaned dataset. The evaluation metrics confirm strong generalization and practical viability for production deployment in logistics workflows.



Interpreting the Metrics:

Accuracy (74%)	The model correctly classifies 74 out of 100 orders as either late or on-time. Given the near-equal class distribution and inherent unpredictability of supply chains, 74% represents a strong baseline that significantly outperforms random guessing (50%) and rule-based heuristics.
Precision (79%)	When the model predicts an order will be late, it is correct 79% of the time. High precision is operationally valuable: it minimises false alarms and ensures that when logistics teams receive a late-delivery alert, the signal is reliable — avoiding alert fatigue and wasted intervention resources.
Recall (75%)	The model captures 75% of all actually late orders. The remaining 25% represent missed late deliveries — a manageable false-negative rate for a v1 model, and one that can be reduced through further feature engineering and hyperparameter optimisation.

Business Value Translation: The practical impact of a 74% accurate model operating on a supply chain processing ~172,000 orders annually is substantial. Assuming an average delay-related cost of \$12.15 per order (derived from \$2.1M / 172,765 delayed orders), predictive intervention on even 30% of correctly identified late orders could prevent **~\$270,000 in annual delay-related costs**. As the model is refined and intervention workflows mature, this figure scales proportionally.

■ *Business Case: Predictive intelligence transforms logistics from a reactive cost center to a proactive value driver. With 79% precision, operations teams can trust the model's alerts — routing flagged orders to priority carriers, pre-notifying customers, or adjusting inventory buffers — before service failures occur and customer relationships are damaged.*

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STRATEGIC RECOMMENDATIONS & CONCLUSION

Based on the analytical findings and predictive model outputs, the following strategic recommendations are prioritised by potential business impact and implementation feasibility.

Delivery Alerts

Integrate the Random Forest classifier into the order management system to trigger alerts for each new order. Orders above a configurable threshold (e.g., 72% predicted delay) should trigger automated alerts to logistics coordinators and proactive customer communication. Transition the model from a reporting tool into an operational intervention system.

First Class Shipping Pipeline

The 100% late-delivery rate for First Class shipping is a systemic failure requiring a full audit of First Class carrier SLAs, routing logic, and order eligibility. Conduct a root cause analysis from misclassification, carrier underperformance, or internal process errors. Once the primary cause is resolved and re-validated.

Service for High-Value Segments

Home Office and Consumer segments exhibit the highest delay sensitivity. Implement tiered service allocation — directing Standard Class and Priority Mail toward profitable customer segments, while applying stricter SLA monitoring for high-value segments.

Key Performance KPI Dashboards

Build operational dashboards surfacing delay rates by shipping method and region in near-real-time. Include model prediction accuracy tracking alongside actual performance monitoring. Establish weekly review cadence with logistics leadership to address emerging trends.

Conclusion

This project demonstrates how business intelligence and machine learning can work in concert to transform raw supply chain transactional data into measurable operational value. The analytical journey — from data cleaning and exploratory analysis through bottleneck detection to predictive modeling — reveals a supply chain with strong profitability fundamentals that are being eroded by systematic delivery failures.

The core finding is clear: **54.71% late delivery rate and \$2.1M in delay-related profit leakage are not inevitable — they are addressable.** The shipping mode anomaly (100% First Class delay rate), the payment pipeline bottlenecks, and the regional variance all point to specific, fixable process failures rather than intractable external constraints.

The Random Forest predictive model, achieving **74% accuracy and 79% precision**, provides the operational intelligence layer to act on these insights proactively. Deployed with real-time alerting and continuous retraining, it positions the organization to shift from reactive delay management to predictive logistics excellence — a strategic advantage in an increasingly competitive supply chain landscape.

What Was Done

What Was Found

What To Do

Analysed 172K+ supply chain orders across 23 features with full EDA, bottleneck mapping, and ML modeling	54.71% late delivery rate, \$2.1M loss, First Class 100% delay anomaly, PAYMENT_REVIEW bottleneck	Deploy predictive alerts, fix First Class routing, implement KPI dashboards, tier customer service
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